***TEAM PROJECT 2***

Business Running Case:

Investing in a New In-House Microbrewery?

***Team 7:***

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## Executive Summary

Buttermilk & Bourbon, a Southern cuisine restaurant in Boston's Back Bay, is evaluating the establishment of an in-house microbrewery against the backdrop of a saturated craft beer market, rising operating costs and stringent regulatory requirements. Despite a maturing market, our prior analysis demonstrates that a microbrewery remains a viable and compelling opportunity through pursuing differentiation with a distinctive food-and-beer pairing concept and targeted marketing strategies. These elements aligned with Buttermilk & Bourbon's goals to strengthen competitive position, elevate the customer experience, and unlock new revenue streams. Building on that foundation, we determine the feasibility of this venture by presenting strategic insights and actionable recommendations grounded in innovation, organizational design, and operations management.

**Key Findings:**

- Innovation & Technology Management - A PESTEL and SWOT analysis shows that the microbrewery is well positioned to meet evolving customer expectations and counter market threats by pursuing two complementary innovation paths, product-driven innovation through a non-alcoholic craft beer line and sustainability-driven innovation through a zero-waste culinary and brewing system, which together strengthens brand differentiation and sustains long-term competitive advantage.
- Organization & HR Management - A lean, hierarchical mechanical organisation is the best fit for running the microbrewery. Our risk analysis identified the loss of key personnel and burnout as the most significant threats, mitigated by focusing on cross training, promoting employee well being and competitive compensation, including a

wage premium of 10-15% for critical employees. Cost benefit analysis for marketing shows that outsourcing offers lower costs (as low as $35,000-$75,000 as compared to $76,000-$102,000) and better benefits, which supports outsourcing the digital media and marketing team as it creates maximum impact while staying safely in the budget.

- Operation Management - Our Sensitivity Analysis and SWOT Analysis revealed that maintaining operational performance within the target excess demand range requires careful adjustment of Percent Monthly Demand allocations. The analysis showed that even small misalignments between demand and supply can result in significant shortages or surpluses across SKUs. By identifying where production consistently fell short or exceeded demand, we confirmed the need for targeted, data-driven adjustments rather than broad capacity changes. This finding strengthens our operations strategy by highlighting the importance of precise forecasting and demand planning to sustain efficiency and support long-term profitability.
- Simulations - Running simulations on this business case shows that the microbrewery's strongest and most sustainable performance emerges when decisions across marketing, finance, operations, organization, and innovation are integrated, as demonstrated in Cycle 14, where coordinated capacity planning, team structure, and innovations across products and sustainability deliver higher efficiency, lower break-even revenue, and a more resilient long-term profitability outlook compared to narrower finance or operations-focused cycles.
- Key financials as seen in Cycle 14: Gross profit $2,664,210.32, net profit $1,936,081.64, break-even point 14.19 months, ROI 111.64% (return of $1.11 for every $1 invested).

- Recommendation - We recommend that Buttermilk & Bourbon should invest in the in-house microbrewery, as the project is strategically and financially feasible and strengthens the restaurant's brand identity through premium pricing, unique curated pairings, sustainability-driven innovation, and targeted marketing.

## Managerial Decision-Making Process

### *Innovation & Technology Management (Sustainability-Driven Innovation)*

In evaluating innovation opportunities for the in-house microbrewery, the most strategically valuable and operationally feasible path for Buttermilk & Bourbon is the implementation of a **Zero-Waste Culinary & Brewing System**. This sustainability-driven innovation aligns directly with the restaurant's Southern culinary identity, strengthens its competitive differentiation in Back Bay, and enhances both efficiency and guest experience without the need for complex technological investments.

The core concept involves a process innovation that transforms the microbrewery into a **closed-loop system** where byproducts or leftovers from brewing, such as spent grains, are repurposed into new culinary applications. These grains can be incorporated into cornbread, biscuit dough blends, dessert crusts, or even used as coating for fried menu items. This innovation deepens the restaurant's eco-culinary brand story and creates a cohesive "farm-to-glass, farm-to-table" experience, linking the brewery and kitchen in a way that no competitor in Back Bay currently offers. Importantly, this system exploits economies of scope, creates cost containment opportunities, and reinforces Buttermilk & Bourbon's identity as a chef-driven Southern restaurant with an emphasis on comfort, creativity, and authenticity.

Choosing this sustainability-focused innovation also reflects a realistic assessment of operational constraints. The microbrewery's current infrastructure supports brewing but not large-scale product diversification. Therefore, an innovation that operates within existing capacity while increasing value through repurposing, process efficiency, and guest experience provides the best return without introducing unnecessary operational complexity. The approach is immediately implementable, requires no additional equipment, and integrates naturally into the restaurant's existing processes.

**Horizon 1: Strengthening Current Operations Through Zero-Waste Integration**

In the short term, the microbrewery focuses on stabilizing core brewing processes while incorporating sustainability practices directly into operations. This includes establishing consistent methods for capturing, storing, and repurposing spent grains into menu items, as well as aligning brewing schedules with kitchen needs. Horizon 1 creates immediate efficiency gains by reducing ingredient wastage and procurement costs, while reinforcing the restaurant's Southern culinary identity through complementary food-and-beer interactions. This phase requires minimal financial investment and delivers high operational impact, making it an ideal foundation for innovation.

**Horizon 2: Expanding Sustainability Practices into Brand Differentiation**

Once the Zero-Waste system is functioning reliably, the next phase involves leveraging sustainability as a visible and marketable differentiator. Practices such as sourcing ingredients from New England farms that improve quality while reducing transportation costs, highlighting zero-waste menu items, and integrating sustainability narratives into tasting events and

promotions allow Buttermilk & Bourbon to build a distinctive eco-culinary brand within Back Bay. Horizon 2 shifts sustainability from a back-of-house efficiency measure to a front-of-house experiential element that enhances perceived value and supports premium pricing. This aligns with the Boston dining market, where environmentally conscious consumers show higher willingness to pay for sustainable dining experiences.

**Horizon 3: Long-Term Sustainability Leadership and Expanded Value Creation**

Over the long term, Buttermilk & Bourbon can aspire to become Boston's leading Southern-inspired eco-culinary brewery, integrating sustainability across all aspects of production, dining experience, and supply chain. This may include developing seasonal zero-waste tasting menus and investing in energy-efficient brewing and water recapture technologies. In Horizon 3, sustainability is rooted in the restaurant's brand identity to preserve competitive advantage and set the foundation for further shared value creation through educational events and long-term community engagement. This sustained positioning ensures differentiation in a mature craft beer and dining market.

## *Innovation & Technology Management (Product-Driven Innovation)*

Using the three horizons framework we explore a way for Buttermilk & Bourbon to balance innovation efforts when launching Southern-flavored non-alcoholic (NA) craft beers with current and future opportunities for growth (Balen, A. V. (n.d.)).

**Horizon 1: Strengthening Current Craft Beer Offering**

Upon optimizing its existing alcoholic craft beer lineup, Buttermilk & Bourbon can engage in product diversification by introducing NA beers as an extension of the core brewing process,

allowing the microbrewery to enhance its current value proposition without disrupting core operations and preventing revenue leakage from nonconsumers or customers seeking NA options elsewhere. NA beers intentionally curated to harmonize with smoked, spicy, and buttery Southern flavors will serve as an indulgent beverage that boasts bold flavours, premium ingredients and wellness advantages. This will allow Buttermilk & Bourbon to differentiate sharply within Boston's evolving craft beer landscape and strengthen both economic and social value creation.

**Horizon 2: Scaling the NA Craft Beer Line (Emerging Opportunities)**

NA beer demand in the U.S. is accelerating rapidly, growing 22% annually from 2019 to 2023 and Massachusetts in particular is emerging as one of the fastest-expanding markets, ranking 3rd amongst the top NA beer drinking states (IWSR, 2024). Developing a lineup of Southern-rooted NA beers that feature ingredients like chicory and molasses & toasted spices to deliver bold, regional flavors not found in mainstream NA beers, positions Buttermilk & Bourbon to capture this market early. Competitor analysis indicates that most restaurants in Back Bay serve canned or bottled NA beers from mainstream brands like Athletic Brewing and Guinness instead of brewing them in-house. As a result, NA options often feel like an afterthought or merely a functional alternative instead of curated beverages designed to complement the restaurant's flavors and dining experience. Hence, Buttermilk & Bourbon can carve a niche through serving fresh, flavorful and exclusive NA craft beers. At no additional cost, the microbrewery can leverage the existing tank system to produce both the alcoholic and non alcoholic beers by adapting the traditional brewing process through controlled or arrested fermentation which stops the alcohol production early. This way, the innovation fits seamlessly into current operational

workflows, requiring no new equipment or major process changes. This technique also preserves the natural flavors and aromas produced during fermentation for a fuller, more complex flavor while keeping alcohol by volume (ABV) below 0.5%, unlike other methods of NA beer production such as post-fermentation de-alcoholization (RahrBSG, 2025).

**Horizon 3: Pioneering Culinary-Driven NA Innovation**

A future-forward vision is one where Buttermilk & Bourbon's microbrewery can lead Boston's next wave of premium NA craft beers by building a new core competency in NA beers to protect and extend its current market position. This can be achieved by developing gastronomic, chef-designed NA beer pairings that rival alcoholic craft beers in complexity. Key differentiators include small touches like serving the NA beers in identical glassware as alcoholic beers, including them on tasting flights and giving full pairing notes when suggesting complementary menu items will create a fully-integrated and holistic dining experience for all.

To successfully commercialize this as a well-curated, flagship product, we will position our NA craft beers as a premium pairing experience pricing it at $8.50 a pint, and targeting sober-curious diners, lunchtime patrons, and corporate groups actively seeking sophisticated and handcrafted non-alcoholic options. This customer segment will be willing to pay higher for quality and flavor. Since total material cost for acquiring the chicory, spices and molasses for these beers is ~$1.00 per unit, this price point also achieves a decent gross margin, making a premium price point financially sound.

## *Organization & HR Management*

At Buttermilk & Bourbon, we plan to follow a lean **mechanical organization structure**, focusing on high quality operations and productivity while being financially efficient. The

**structure is decidedly hierarchical** as it provides us with clear lines of authority starting from the owner, all the way down to the specialized staff. This helps us ensure deep expertise and accountability for all tasks which in turn, is better for product quality and efficient distribution. The **centralized top down decision making** in this structure provides clear command and control ensuring stability and consistency. While this structure offers advantages in efficiency and cost management, we do recognise that it may result in slower decision making especially in areas like innovation which is why we plan to hire a dedicated product innovation specialist after the first financial year. We will have a core team of **8 employees** allowing us to cover all core functions without any unnecessary overheads. Please refer to the appendix figure 1 for the organizational chart.

We integrate this structure with our overall strategy by ensuring that our team drives productivity and financial performance, supported by our specialized roles which are critical for our quality and distribution relationships. Through our **opportunity analysis** we realised a need for highly **competitive compensation** like a 10-15% wage premium for our brewmaster and an intentionally high pay for our salesperson to be able to retain the top talent in the Boston market. This salary compensation approach will help us prevent turnover.

According to our **three year plan**, we will establish the core team in year 1. In year 2, we will expand our team by hiring a product innovation specialist to help us with R&D initiatives and drive the NA beers development. We will also implement the first planned wage adjustments. By year 3, we plan to implement the second round of wage adjustments and focus on stabilizing the new R&D functions while also  assessing the need for further expansion based on our revenue and growth.

For administrative tasks, we will be employing one **part time office assistant** assigned at an annual salary of **$30,000**. They will be handling all day to day office coordination and will be the main point of contact with our external digital marketing team. This salary falls slightly below the Boston average for entry level roles, however it is justified as this is a part time role and does not require any prior technical skills. Once the reduced working hours are adjusted, the salary remains competitive.

Our wholesale distribution partner will be operating under a separate contractual agreement. For smoother operations and optimal communication, we will be hiring a **salesman** at **$70,000.** This compensation is intentionally set 8-10% higher than the median recognizing the role's importance and to strengthen the long term retention of the employee.

Our **Brewmaster** will be compensated at **$100,000** and they will be overseeing the microbrewery system, including the craft beer production, quality and flavour assurance and recipe formulation. According to metropolitan areas, this salary aligns with the upper mid tier benchmarks for brewmasters. Due to  a limited number of experienced craft beer professionals and the higher operating costs in Boston's Back Bay area, a 10-15% wage premium is commanded over the national average for brewing specialists, which justifies this figure.

Additionally, our staffing plan includes a team of specialized **consultants** that are compensated at **$90,000** annually reflecting the market rates for fractional consulting arrangements. In Boston, consultants usually price their services based on expertise rather than hours and this price point reflects the midpoint for ongoing advising roles in the F&B sector.

Starting year 2, we plan to extend our team by hiring a **product innovation specialist** with an annual salary of **$85,000**. Their role will be focused on developing the new flavored

non-alcoholic beers, assisting the Brewmaster and supporting future R&D initiatives to ensure our offerings remain competitive and aligned to emerging consumer trends. The salary for the role is justified by the specialized expertise required for the role. Additionally, salaries for product development tend to be 10-12% above the national average in Boston, reflecting the high competition and cost of living in the area. Offering a competitive salary is essential to attract qualified candidates as this position plays a direct role in driving our long term product differentiation and revenue growth.

In order to support employee satisfaction and prevent turnover risks, we have incorporated planned wage adjustments in our budget. We will be offering a **10%** annual increase for our **salesperson** due to the strategic importance of maintaining distributor relationships. For the **administrative and executive roles**, we will be providing a **5%** annual growth which aligns with the average wage inflation forecasts for the Boston food service sector. We also assume a **5%** salary increase in year 3 for our **product innovation specialist**. This number is also consistent with the wage inflation for mid level product development professionals.

## *Operations Management & Decision Making*

In developing our operations management strategy for the microbrewery, we relied heavily on the operations planning framework to refine our production and capacity decisions. One of the key operational adjustments we made was shifting the cutoff point for sales and production planning, ensuring that brewing cycles aligned more accurately with actual demand patterns and minimized excess inventory. We also incorporated seasonal forecasting in the percent monthly demand to adjust our sales  projections based on predictable fluctuations in beer consumption. Industry research shows that beer demand typically peaks during warmer months (May - Sep)

and declines in winter (Oct - March) , which directly affects capacity planning, ingredient purchasing, staffing, and tank utilization (Brewers Association, 2024). By integrating seasonality into our demand forecasts, we were able to model more realistic production volumes and design a brewing schedule that reduces operational inefficiencies such as overproduction or stock shortages. Most importantly, these adjustments enabled us to maintain an excess demand range between –10% and 10%, ensuring that our production plans stay closely aligned with actual customer demand. Together, these operations management decisions strengthened the overall feasibility of our microbrewery by supporting more accurate planning, reduced waste, and a more efficient allocation of resources.

**Integration of the Operational Strategy with Company Strategy and Resources**

The operational strategy for the microbrewery directly supports Buttermilk & Bourbon's broader strategic objective of differentiating through unique, experience-based offerings and strengthening its competitive position in the Back Bay market. Rothaermel (2025) emphasizes that operational capabilities within the value chain such as production efficiency, process design, and quality management are foundational to sustaining competitive advantage. In alignment with this perspective, our operations plan focuses on brewing capacity, batch scheduling, and quality control to ensure that production activities reinforce the restaurant's strategic positioning around freshness, exclusivity, and experiential dining.

Using the operations planning framework, we refined production and capacity decisions by shifting the cutoff point for sales and production planning to better synchronize brewing cycles with actual customer demand. Seasonal forecasting was incorporated as a strategy to model

predictable fluctuations in beer consumption, consistent with Rothaermel's emphasis on data-driven planning and adaptation to external conditions. Integrating craft beer product seasonality through adjusting allowed us to maintain an excess demand range between –10% and 10%, reduce waste, allocate resources more effectively, and support a stronger overall value chain contribution (Rothaermel, 2025).

**Opportunity Analysis**

The microbrewery business model presents a strong strategic opportunity by enhancing the restaurant's differentiation within an otherwise saturated dining market. According to Rothaermel (2025), firms achieve competitive advantage when they build capabilities that are valuable, rare, difficult to imitate, and organizationally supported elements foundational to the VRIO framework. The microbrewery strengthens these capabilities by enabling on-site/in-house, small-batch production that its competitors in the Back Bay area do not widely offer.

Operational core competencies include rapid responsiveness to demand, the ability to update seasonal brews, and integrating brewing processes into the restaurant's brand storytelling, capabilities that enhance both customer value and strategic positioning. Our previous competitor analysis in TTP 1, showed limited availability of in-house craft beer experiences specifically with the pairing of southern cuisine nearby,  creating a distinct niche and potential opportunity for Buttermilk & Bourbon. Operational differentiators such as rotating seasonal brews, integrating non alcoholic beers into the system, visible brewing processes, and highly responsive production cycles can further reinforce Buttermilk & Bourbon's ability to capture and sustain customer loyalty through a unique value proposition.

**Developing Three-Year Plans for Successful Operations**

The first year of operations focuses on establishing the operational foundation required for the microbrewery to function effectively. This includes installing all brewing equipment, securing necessary licenses, and ensuring compliance with safety and health regulations. Standard Operating Procedures (SOPs) are developed for brewing, sanitation, tank scheduling, and quality control, creating the structured processes that Rothaermel (2025) identifies as essential to strengthening value chain activities. Staff hiring and training also takes place during this initial year, with an emphasis on integrating demand forecasting into weekly production planning. The year concludes with the launch of flagship brews and pilot production runs, which help establish baseline yields, cycle times, and initial performance benchmarks. Additionally, the microbrewery will take on small-scale distribution opportunities to extend its market reach, including wholesale production to nearby restaurants, specialty stores, or local partners.

In the second year, the focus shifts toward optimization and capability building. Brewing schedules are updated using historical sales data and seasonal forecasting to better align production with customer demand. Seasonal brews are introduced to enhance differentiation and reinforce the Buttermilk & Bourbon's strategic positioning. Efficiency improvements are pursued through lean operations principles including reducing waste, improving flow, and refining tank utilization, which reflects Rothaermel's (2025) emphasis on process improvements as a source of competitive strength. During this phase, coordination between marketing and operations becomes increasingly important to ensure that production capacity matches promotional campaigns and customer traffic patterns.

By the third year, the microbrewery is positioned to scale its operations and strengthen its strategic impact within the restaurant. Decision-support tools, such as sensitivity analysis and what-if models, are implemented to further refine capacity planning and demand fulfillment. Supply chain resilience is enhanced through diversified vendor relationships to ensure ingredient availability and reduce operational risk. This phased development plan reflects Rothaermel's (2025) view that organizations must align internal capabilities with long-term strategic objectives while remaining adaptive to changing market conditions.

**Implementation, Management, and Control of the Three-Year Program**

Implementation is supported through structured performance measurement systems that ensure operations remain aligned with strategic goals. Consistent with Rothaermel's (2025) value chain perspective, operational controls focus on improving efficiency, quality, and responsiveness. Key performance indicators include batch yields, brewing cycle time, waste percentage, cost per batch, demand fulfillment accuracy, and equipment utilization.

Quality control will be maintained through routine batch testing, adherence to SOPs, and consistent sanitation standards. Monthly operational reviews and continuous improvement initiatives grounded in lean principles ensure that the microbrewery remains agile and capable of adapting to fluctuations in demand or resource constraints. This control system strengthens the microbrewery's contribution to competitive advantage by reinforcing reliability, reducing inefficiencies, and supporting consistent delivery of customer value.

## Application of Decision Support Tools

### *Innovation & Technology Management*

**PESTEL Analysis**

The sociocultural factor from our PESTEL analysis revealed notable shifts in Boston's drinking culture, particularly the rise of the "sober-curious" movement and growing health-conscious drinking habits (Cooke, 2025) which is more evident among the younger demographic, where 61% of Gen Z and Millennial consumers now prefer low-to-non-alcoholic options (Gribbins, 2025). Boston's dining market also includes many working professionals, tourists, and lunch-time diners who may prefer NA beverages for mid-day dining or corporate events. Having an in-house microbrewery enables the management team to quickly capitalize on these social trends, to directly capture emerging market segments and bring in nonconsumers, making the brand more inclusive. This is especially valuable in an environment where tariff-driven price pressures may otherwise dampen demand for alcoholic offerings. NA craft beer allows revenue growth without increasing exposure to excise taxes or alcohol-related operational costs. Environmentally, NA beers often require less energy because fermentation can be halted early (Smith, 2024), bolstering our eco-culinary brand story. These are the main drivers for innovating a dedicated flavorful non-alcoholic craft beer line within our microbrewery.

**SWOT Analysis**

To evaluate the feasibility of the Zero-Waste Culinary & Brewing System, a SWOT analysis was conducted to assess internal capabilities and external conditions driving the innovation. The strengths of this innovation lie primarily in its alignment with Buttermilk & Bourbon's existing brand identity and operational processes. As a chef-driven Southern restaurant, the ability to

integrate spent grains into comfort-focused dishes enhances menu richness while reducing waste. The restaurant also benefits from growing consumer demand for sustainable dining experiences, which increases willingness to pay and supports premium pricing. These factors collectively position sustainability as a powerful differentiator in a competitive market.

However, the innovation also presents internal weaknesses that require careful management. Implementing a zero-waste system demands coordination between the brewery and kitchen, as well as consistent staff training to ensure proper handling and repurposing of ingredients. Dependence on local suppliers introduces potential seasonality challenges, and the financial benefits of sustainability may take time to fully materialize. These internal considerations highlight the importance of creating standardized processes and cross-functional communication systems as part of the implementation plan.

Externally, the innovation provides multiple opportunities for brand growth and market differentiation. Boston's consumer base is increasingly eco-conscious, with many diners actively seeking restaurants that demonstrate environmental responsibility. This trend, combined with the scarcity of sustainability-focused breweries in Back Bay, creates a strong opportunity for Buttermilk & Bourbon to define a new market niche as an eco-culinary Southern brewery. Additionally, the innovation enhances public relations potential, community partnerships, and visibility in local events and festivals.

Threats primarily originate from external factors such as competitors adopting similar messaging without implementing genuine practices (greenwashing), supply chain variability among local producers, and tightening regulatory guidelines related to food repurposing or sustainability

claims. These potential threats reinforce the need for transparency, quality control, and strong supplier relationships to maintain authenticity and operational continuity.

Overall, the SWOT analysis confirms that the Zero-Waste Culinary & Brewing System is not only feasible but strategically advantageous. It offers strong alignment with brand values, operational capabilities, and market trends, while presenting manageable risks that can be mitigated through effective planning and coordination.

## *Organization & HR Management*

**Risk Analysis**

To assess the vulnerability within our organizational and HR structure, we conducted a risk analysis matrix using a probability impact system. By multiplying likelihood and potential impact, we were able to figure out the risks that need immediate action. The factors with higher scores pose the most critical threat to our brewery. Refer to the appendix figure 2 and 3 for better reference.

We evaluated four such risks:

1) ***Loss of key personnel***

   This is our biggest concern with a risk score of 25. Losing any essential employees such as our brewmaster or innovation manager would create disruption in our systems and cause inconsistencies in our performance. This could delay production, halt recipe development and negatively impact our deadlines.

   To mitigate these risks, we must implement an approach focused on knowledge preservation and talent stability. We will be cross training employees on core brewing processes and strictly rely on documentation of all recipes to reduce the dependence on a

singular person. We are offering our employees competitive pay as a part of our retention efforts.

2) ***Skill Gap***

As we scale our microbrewery, there may arise a situation where our existing staff do not have the necessary technical or operational knowledge for more advanced brewing such as the arrested fermentation technique to support our NA beer innovation, and quality control. This may create a risk that can slow innovation and reduce competitiveness. While this risk has a score of 12, we still need to keep some strategies in mind to be able to counter it. As such, we will have mandatory training workshops focusing on brewing science, safety and equipment use on regular intervals for all our employees. We will conduct skill assessments to keep track of emerging gaps early on to be able to step in and prevent it. In the future, we will also be implementing mentorship structures where senior employees can guide the other staff.

3) ***Burnout***

As we are already working with a lean team, times of high workloads may create undue pressure on our employees and they may burnout leading to absenteeism, lower productivity and morale. This is the second biggest threat to us with a risk score of 15. Introducing rotating responsibilities and schedules will provide relief for our employees in such periods. We plan to promote a culture of well being by encouraging timely breaks, and monitoring overtime hours which will ensure enhanced employee engagement through increased job satisfaction and motivation.

4) ***Inability to offer competitive wages***

Our ability to retain key employees rests on offering competitive wages. If the salaries fall short of the industry standards, there are higher chances of employee turnover. This stands at a risk score of 9 which is considered moderate, however, we should still keep an eye on it due to the highly competitive nature of Boston's food and beverage industry.

To stay on top of this, we will conduct annual salary benchmarking to stay at par with the industry standards. Developing benefits beyond monetary value, like staff meal programs and profit sharing models will give us an edge and create a deeper bond with the employees. More hands-on involvement like staff tasting sessions and involvement in innovation will also make the team more close knit.

***Decision implications***

This risk analysis has shaped our operational decisions for the microbrewery. The high risk score for loss of key personnel has made us prioritise talent retention and cross training as foundational strategies. Similarly the other moderate and high risks like skill gap and burnout indicate that the microbrewery requires a very structured workforce plan including training and documentation. The wage risk influenced our budgeting decisions, making sure that compensation for all employees remained competitive.

**Cost-Benefit Analysis**

The purpose of this Cost-Benefit analysis is to determine whether we should keep an in-house social media and marketing team or outsource them from a professional digital media agency. We have allocated a budget of $150,000 for all marketing activities, including social media, campaign execution, partnerships and brand building. In this analysis, we assess the **direct, indirect and opportunity costs** as well as **tangible and intangible benefits** associated with both these possible methods to be able to choose the alternative that is more financially sound

and sustainable. All the assumptions are within the current industry standards. For a detailed breakdown of all the tables, please refer to figures 4 to 8 of the appendix.

Summary Table

| **Category** | **In-House team** | **Outsourced** |
|---|---|---|
| Annual Cost | $91,000-$126,000 | $37,000-$73,000 |
| Indirect Costs | High | Low |
| Opportunity Costs | High | Low |
| Tangible Benefits | Integration & Availability | Higher Performance overall |
| Intangible Benefits | Seamless Cultural Fit | Larger Skillset & Innovation |
| Impact on Budget | Fits under the budget | Fits under the budget with surplus |

For the **direct costs,** we are assuming 1 employee for our team and it is their salary that we have added as a number. Ideally there is only one direct cost related to the outsourced agency, which would be the fee. In case of adversity, we are also assuming a surplus amount for extra campaigns. For the in-house team, we have salaries, overheads, software and office needs to consider which for the outsourced team is all inclusive.

The **indirect costs** are higher for the In-house team as HR spends a lot of time trying to find the best recruits for the company and even after that, the standard onboarding cost per employee is $1200-2000. Even conservatively, these costs add up to at least $40-45000 for an in-house team while the costs are minimal ($5-10,000) for the outsourced team. The only spend is that in management oversight which is also only about 3-4 hours weekly.

The **opportunity costs** essentially refer to the economic value of the choices we make. Realistically, for the in-house team, the productivity will be much lower early on as compared to the outsourced agency. We will have a limited skill set and longer campaign development cycles and a much higher chance of turnover which is negligible with an outsourced team. The brand also faces potential risk of loss of revenue due to slower execution. Thus, the opportunity cost of an in-house team is $25-30,00 while it is only $5-10,000 for an outsourced agency.

**Tangible and Intangible Benefits**

The outsourced agency gives us more tangible and intangible benefits as it is a collection of more specialized services and personnel all under one major cost as compared to the in-house option wherein we will have to invest and reinvest for different functions and skills. Being a team of many people with various technical skills, the agency also gives us more content and creative output. We also face a much higher risk of turnover with an in-house team which in turn also creates a big impact on the overall project in case someone from the team decides to leave. This risk is almost negligible when it comes to an outsourced agency as the agency bears the impact of leave of personnel. In terms of cultural alignment and integration, the in-house team is a better option however the overall benefits from the outsourced agency outweigh that of the in-house team.

## *Operations Management*

*Refer to Appendix Figure 9 and 10 for supporting chart data.*

**Sensitivity Analysis + SWOT Analysis**

**Rationale for Using Prior Fiscal Years as Baselines for Subsequent Planning**

FY-1 is used as the baseline because it provides the first full year of actual operational data, showing how demand, supply, and capacity behaved under real conditions. Sensitivity analysis applied to FY-1 allows us to identify shortages, surpluses, and SKU responsiveness, which then informs more accurate capacity and demand-planning decisions for FY-2. In turn, FY-2 becomes the next baseline for FY-3, creating a logical year-over-year improvement cycle.

**Findings of The Sensitivity Analysis**

The sensitivity analysis showed that each SKU responds differently to changes in production, meaning the brewery cannot rely on a single adjustment across all products. BR-1 and BR-2 consistently showed moderate shortages that worsened when production decreased and improved when production increased, indicating they require slightly higher demand allocation. In contrast, BR-3, BR-9, and BR-10 showed persistent surpluses and moved deeper into excess supply when production increased, meaning they needed reduced Percent Monthly Demand. BR-4 displayed the highest volatility, with small adjustments causing large swings, suggesting it must be modified cautiously.

More stable SKUs, such as BR-5, BR-7, BR-8, and BR-6, showed predictable responses to the ±10% scenarios, meaning only small changes are needed to correct their imbalances. Overall, the findings confirm that production alignment issues are SKU-specific, and optimal performance requires targeted PMD adjustments for each product rather than broad system-wide changes.

## SWOT ANALYSIS

When the SWOT analysis is viewed alongside the sensitivity findings, the strategic implications of the numerical results become much clearer. The sensitivity analysis showed that several SKUs including BR-1 and BR-2 experienced consistent shortages at baseline (+5–6% ED) and became severely misaligned when supply decreased (up to +15–16% ED). These patterns highlight an internal strength in demand potential for these product lines, suggesting strong customer appetite for the microbrewery's offerings. At the same time, the data reflect an operational weakness, as current production levels are not fully meeting demand, reinforcing the need for improved forecasting and PMD allocation.

Similarly, the pronounced surpluses in SKUs such as BR-3, BR-9, and BR-10 (baseline ED around –9.5%, dropping to –19% with a 10% production increase) directly support the SWOT finding that the microbrewery faces the weakness of overproducing certain styles. This surplus behavior, visualized clearly in the sensitivity chart, aligns with the threat of operational inefficiency since carrying excess inventory increases costs, ties up tank capacity, and may reduce profitability if products expire or require discounting.

On the opportunity side, the numbers illustrate the microbrewery's potential for strategic improvement. SKUs with moderate shortages (BR-1, BR-2, BR-7, BR-6) demonstrate that small PMD increases could bring them into the –10% to +10% optimal range, representing a clear opportunity to boost service levels and capitalize on strong demand. Conversely, surplus-prone SKUs offer an opportunity to reduce waste and reallocate capacity toward top-selling items. Together, the charts show how targeted PMD adjustments for each SKU can operationalize the broader opportunities identified in SWOT.

Finally, BR-4's volatility, swinging from –8% to –17% surplus with a +10% increase, illustrates a strategic threat identified in the SWOT: operational instability. The chart highlights how even small production changes can destabilize performance for certain beers, emphasizing the need for cautious, data-driven adjustments.

In summary, the quantitative results in the sensitivity figure provide the operational detail that validates the SWOT analysis. The data confirms which SKUs represent strengths (stable performers), weaknesses (misaligned supply), opportunities (demand-responsive products), and threats (volatile or overproduced items). Integrating both tools offers a more complete view of how the microbrewery can align production with demand and position itself competitively within the broader restaurant business.

**Our Business Decision**

The sensitivity analysis was used to examine how changes in Percent Monthly Demand (PMD) affect shortages, surpluses, and overall alignment between demand and supply. This tool was selected because it clearly shows how small PMD adjustments influence Excess Demand and allows us to identify which SKUs consistently fall outside the target performance range of –10% to +10%. The findings were further interpreted through a SWOT analysis, which highlighted which imbalances represented operational weaknesses, which stable products reflected strengths, and where opportunities existed to improve efficiency.

Based on these results, the operational decision we made was to adjust PMD values at the individual SKU level. Shortage-prone SKUs (such as BR-1, BR-2, BR-6, and BR-7) require slightly higher PMD allocations to better meet demand, while surplus-prone SKUs (such as

BR-3, BR-9, and BR-10) require reduced PMD to prevent overproduction. This targeted PMD adjustment strategy aligns with both the sensitivity results and the SWOT insights, ensuring each product moves closer to the desired Excess Demand range while maintaining its seasonal pattern.

## Simulation Cycles

We have simulated decision-making on areas such as finance, marketing, operations, organization and innovation to evaluate their impact on the projected performance of the microbrewery at Buttermilk & Bourbon. The best results were seen in Cycle 8 - which took into account marketing, finance and operational decisions, Cycle 12 - which took into account marketing and finance decisions, and Cycle 14 - which took into account marketing, finance, operations, organization and innovation decisions. Below is an overview of the key financials of these three cycles:

| | **Cycle 8 (Operation)** | **Cycle 12 (Finance)** | **Cycle 14 (Operation)** |
|---|---|---|---|
| Total Revenue | $ 5,413,158 | $ 4,732,398 | $ 5,333,798 |
| Gross Profit | $ 2,705,582.22 | $ 1,911,707.82 | $ 2,664,210.32 |
| Net Profit | $ 1,966,146.6 | $ 1,389,238.07 | $ 1,936,081.64 |
| Break-even Time (Months) | 14.27 | 18.21 | 14.19 |

Cycle 12 provides the narrowest financial-only view, focusing primarily on advertising decisions made to promote the 6 alcoholic craft beers brewed in-house. It gives us an understanding of how changes in marketing spend across efforts such as local events and sponsorships, tasting events and digital marketing together with administrative, employee and rent expenses affect financial outcomes like revenue and profit of the microbrewery business. While marketing

generates demand for the craft beers and is one of the main functions of the microbrewery, without synergy with operations, organization and innovation we saw it generated the lowest revenue ($4.73M), lowest gross profit ($1.91M), and lowest net profit ($651K). Although Cycle 12 appears efficient with a Profit Efficiency Ratio of 18.21, just focussing on this would be misleading because the cycle excludes operational constraints, organizational effectiveness, and product strategy in terms of innovation efforts which are all factors that are essential to maximize the potential of the microbrewery. Without the ability to adjust capacity planning, waste levels, or team coordination, Cycle 12 lacks the levers required to drive meaningful long-term business growth.

Cycle 8 introduces operational decisions such as production planning, impact of materials and labor costs, product pricing, constraint management, and waste control, alongside advertising. This shift improves outcomes with total revenue increasing to $5.41M, gross profit to $2.70M, and net profit nearly doubling to $993K. In this cycle we have simulated decisions such as accounting for seasonal demand variations across the year for the craft beers and adjusting cut off points of the brewery tanks for optimal capacity planning to strike the right balance between satisfying demand while ensuring tanks do not go idle or produce in excess. These improvements highlight how strongly operational efficiency drives profitability in a microbrewery, where batch sizes, brewing seasonality, and waste minimization have direct positive financial implications. Cycle 8 also shows better structural efficiency than Cycle 12, with fixed costs dropping from 0.36 to 0.22 over time and a more favorable break-even point. However, Cycle 8 does not account for organizational dynamics or innovation, which limits its long-term strategic insight.

Cycle 14 delivers the most holistic and realistic assessment. By integrating advertising, operations, organizational coordination, and product innovation (including the launch of two new non-alcoholic beers), Cycle 14 produces a highly balanced performance profile. It generates strong revenue ($5.33M) and high gross profit ($2.66M) while maintaining efficient net profit ($915K). We have made additional key decisions in this cycle on organizational design and innovation opportunities by implementing a mechanical organizational structure that optimized the effectiveness and efficiency of running the microbrewery with clearly defined roles, responsibilities, and flow of information in this new business unit. Another key decision was made on product innovation where added two new Southern-flavored non-alcoholic beers, accounting for their targeted market size and how they fit in with the overall demand of the other 6 alcoholic craft beers. We also made operational decisions for these new products on costs, price, tank cut off point and monthly demand fluctuations to optimize production levels and accounted for hiring a dedicated product innovation developer in FY-2 to drive and sustain this initiative. With these additions, Cycle 14 achieves the lowest break-even revenue ($2.10M) and the strongest improvement in fixed-cost efficiency over time (declining to 0.17), reflecting how innovations, team alignment, and streamlined processes enhance long-term viability. While its profit efficiency is slightly lower than Cycle 12, Cycle 14's results are far more sustainable because they arise from coordinated decisions across production, people, marketing, and product strategy.

Overall, the performance progression from Cycles 12 (finance + marketing) to Cycle 8 (finance + marketing + operation) to Cycle 14 (finance + marketing + operation + organization + innovation) demonstrates that a more comprehensive decision set results in a more accurate and strategically useful performance outlook. For the Buttermilk & Bourbon microbrewery, Cycle 14

provides the strongest foundation for evaluating profitability, managing operational risks, and leveraging innovation to differentiate the restaurant.

## Decision Strategy Summary

The results of our multi-stage analysis clearly demonstrate that the in-house microbrewery offers strong strategic, operational, and financial value for Buttermilk & Bourbon. The integration of innovation (Zero-Waste Culinary & Brewing System and NA beer roadmap), optimized operations planning (cutoff-point adjustments, seasonality, PMD tuning), and a lean organizational structure produced consistently positive outcomes across decision-support tools and BSIM simulations.

Across the simulated cycles, Cycle 14 delivered the most accurate, comprehensive, and strategically useful financial performance, as it integrated all key functional decisions finance, marketing, operations, organization, and innovation. Cycle 14 achieved:

- **Total Revenue:** $5,333,798
- **Gross Profit:** $2,664,210.32
- **Net Profit:** $1,936,081.64
- **Break-even Time:** ~ 14.19 months
- **Improvement in fixed-cost efficiency (declining to 0.17)**

These outcomes indicate that once the microbrewery's production planning, organizational design, and innovation elements are aligned, the project becomes not only profitable but also operationally stable and strategically differentiated.

The Zero-Waste Culinary & Brewing System reduces food waste and procurement costs while enhancing the brand's Southern identity. Meanwhile, the NA beer innovation phased in over Horizons 1–3 expands the consumer base and strengthens long-term competitive advantage. Sensitivity analysis confirmed that PMD adjustments can bring all SKUs within the –10% to +10% excess-demand target, reducing shortages and surpluses. The risk analysis showed manageable vulnerabilities that can be mitigated through structured training, cross-functional communication, and competitive compensation.

Together, these results provide strong evidence that the microbrewery is a high-value investment opportunity.

## *Recommended Strategy*

**Recommendation: The owner should invest in the in-house microbrewery.**

The full integration of marketing, financial decisions, operations planning, organizational structure, and innovation (as captured in Cycle 14) demonstrates that the microbrewery:

1. **Is financially viable**, generating $5.33M in annual revenue and achieving break-even within approximately 14 months.
2. **Leverages the restaurant's strengths**, including its Southern identity, storytelling-based dining experience, and location in a competitive hospitality district.
3. **Differentiates the brand**, especially through the Zero-Waste Culinary & Brewing innovation and future NA beer line.
4. **Reduces long-term costs**, particularly through waste minimization, improved capacity planning, and outsourcing marketing strategically.

5. **Expands market reach**, especially to sober-curious diners, corporate groups, and eco-conscious consumers.

There is **no evidence** in any simulation cycle or decision-support tool suggesting that the owner should decline or postpone the investment. Cycle 14's performance shows clearly that **aligned cross-functional decisions unlock maximum profitability and operational efficiency**.

Thus, the recommended strategy is to **proceed with the investment as designed**, following the implementation plan below.

## *Implementation Plan for the Selected Strategy*

The microbrewery launch should follow a **three-year phased implementation plan**, ensuring operational stability, financial control, and smooth integration of innovation.

### Year 1 - Foundation & Operational Setup

- Install brewing equipment and secure all regulatory licenses.
- Develop SOPs for brewing, sanitation, tank scheduling, and quality control.
- Establish the Zero-Waste system: collection, storage, and repurposing of spent grains.
- Hire and onboard core staff: brewmaster, sales manager, assistant, and consultants.
- Refine PMD values, implement seasonality adjustments, and align brewing schedules with demand forecasts.
- Launch initial lineup of six alcoholic craft beers.
- Begin limited wholesale distribution to test production scalability.

### Year 2 - Optimization & Innovation Expansion

- Integrate historical data into production planning for improved forecasting accuracy.
- Introduce seasonal brews and refine capacity utilization based on FY-1 performance.
- Hire the Product Innovation Specialist to support NA beer development and diversification.
- Pilot the first Southern-inspired NA craft beers using arrested fermentation.
- Strengthen cross-functional communication between kitchen and brewery for Zero-Waste system consistency.
- Improve tank utilization and reduce surplus SKUs using sensitivity-analysis insights.

### Year 3 - Scaling & Strategic Growth

- Expand the NA beer line with new flavor profiles tied to Southern culinary themes.
- Explore energy-efficient brewing technologies and water recapture systems for long-term sustainability.
- Strengthen partnerships with New England suppliers to enhance the farm-to-glass narrative.
- Increase wholesale distribution capacity and evaluate new local retail channels.
- Conduct semi-annual reviews of KPIs such as batch yield, cost per batch, waste %, and ED alignment.
- Host zero-waste tasting events and NA beer pairing experiences to enhance brand visibility.

This staged timeline ensures a realistic, cost-controlled, and strategically aligned rollout of the microbrewery.

## Conclusions

The comprehensive evaluation across innovation, operations, HR, and simulation modeling demonstrates that an in-house microbrewery is not only feasible but highly advantageous for Buttermilk & Bourbon. The project offers a rare combination of financial performance, operational efficiency, brand differentiation, and long-term competitive positioning. The microbrewery strengthens the restaurant's culinary identity, expands its target market, and introduces new revenue streams while maintaining manageable risk levels.

Cycle 14 confirms that when all functional levers are aligned, the microbrewery achieves revenue exceeding $5.3M, strong gross and net profit margins, and quick recovery of initial investment within about 14 months. Beyond financial returns, the Zero-Waste innovation and NA beer roadmap position the restaurant at the forefront of eco-culinary dining and evolving consumer preferences in Boston.

Given the combined insights from decision-support tools, risk analysis, and BSIM performance, the optimal strategic decision is clear:
Buttermilk & Bourbon should invest in the in-house microbrewery and follow the phased implementation plan to ensure successful launch, sustained profitability, and long-term competitive advantage.

# Appendix

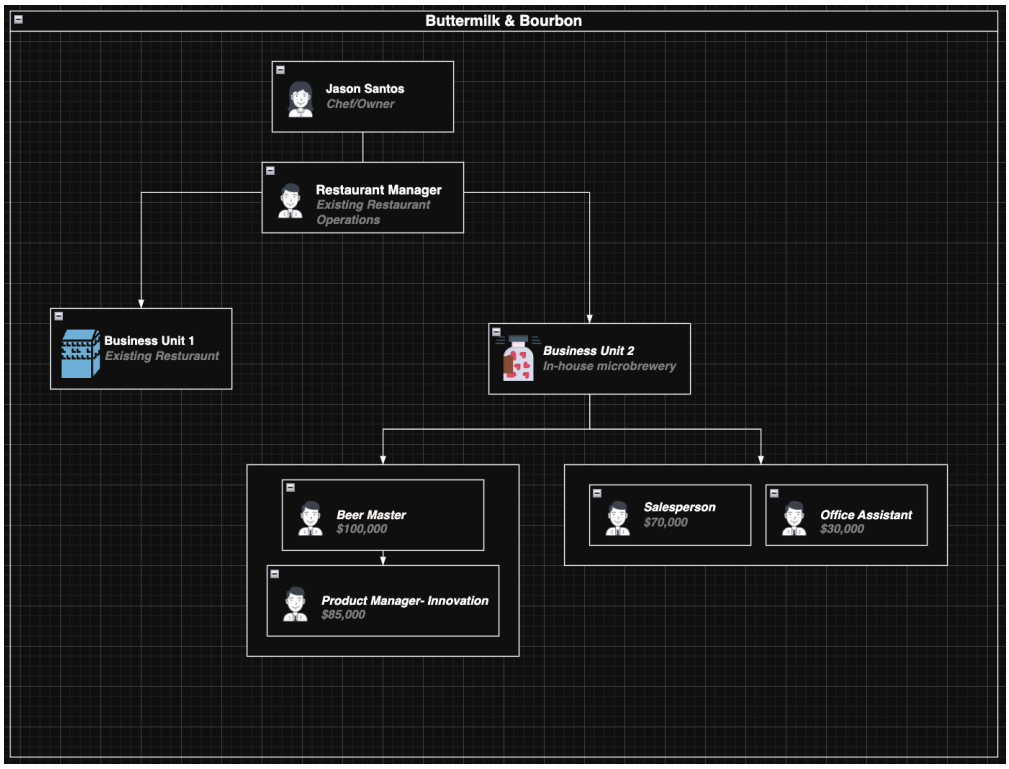


*Figure 1: Organisational Structure*

## *Risk Analysis*

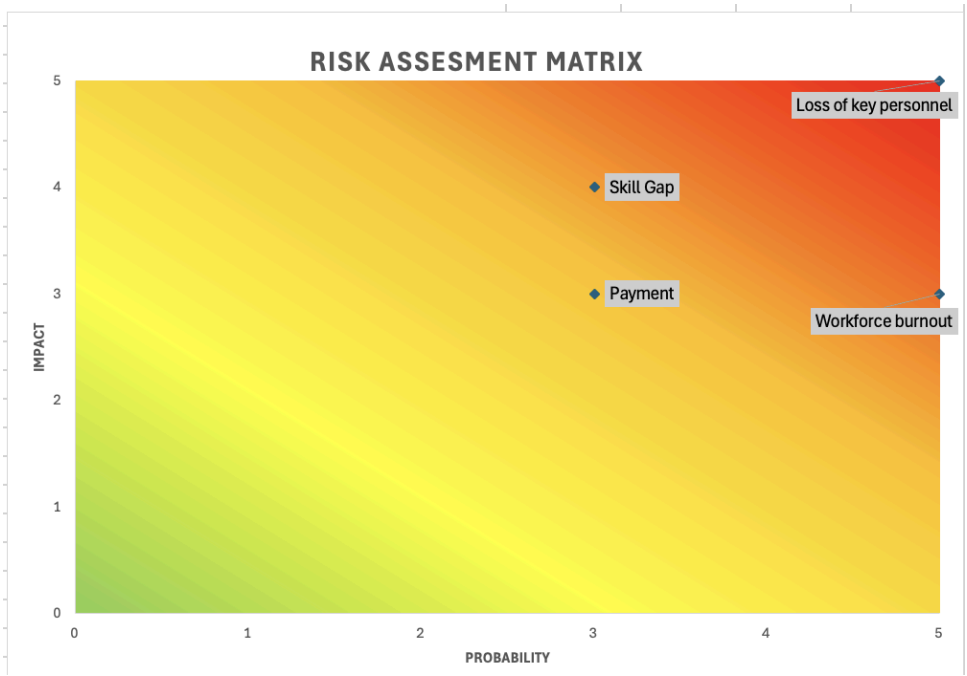

*Figure 2: Risk Analysis Matrix*

| Probability \ Impact | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 5 | 5 | 10 | 15 | 20 | 25 |
| 4 | 4 | 8 | 12 | 16 | 20 |
| 3 | 3 | 6 | 9 | 12 | 15 |
| 2 | 2 | 4 | 6 | 8 | 10 |
| 1 | 1 | 2 | 3 | 4 | 5 |

| Probability \ Impact | Negligable | Minor | Moderate | Major | Critical |
|---|---|---|---|---|---|
| Very Likely | Medium | High | High | Extreme | Extreme |
| Likely | Medium | Medium | High | High | Extreme |
| Possible | Low | Medium | Medium | High | High |
| Unlikely | Low | Low | Medium | Medium | High |
| Rare | Low | Low | Low | Medium | Medium |

*Figure 3: Risk Analysis Key*

### *Cost Benefit Analysis*

Direct Costs Table

| **Category** | **In-house Marketing team** | **Outsourced Agency** |
|---|---|---|
| Salary | 55,000-70,000 | N/A |
| Benefits and Overheads | 15,000-20,000 | N/A |
| Software and Tools | 6000-12000 | Included |
| Agency Fee (Retainer) | N/A | 30,000-60,000 |
| Extra Creative projects | N/A | 5,000-10,000 |
| Total | 76,000-102,000 | 35,000-75000 |

*Figure 4: Direct Costs for CBA*

Indirect Costs Table

| **Category** | **In-house Marketing team** | **Outsourced Agency** |
|---|---|---|
| Training & Skill upgrade | $3,000-5000 | None |

| | | |
|---|---|---|
| Turnover Risk | High replacement Cost $5,000-8,000 | None |
| Management Supervision Time | Moderate 5-10 Hours | Minimal time |

*Figure 5: Indirect Costs for CBA*

Opportunity Cost Table

| **Opportunity Cost** | **In-house Marketing team** | **Outsourced Agency** |
|---|---|---|
| Hiring and Onboarding time | High | None |
| Diversity of expertise | High | Low |
| Lost productivity during time off | High | Low |

*Figure 6: Opportunity Costs for CBA*

***Benefit Breakdown***

Tangible Benefits

| **Benefit** | **In-House** | **Outsourced** |
|---|---|---|
| Availability (Full time) | Strong | Medium |
| Integration (cross functionality) | Strong | Medium |
| Creative Output | Limited | High |
| Content Volume | Moderate | High |

*Figure 7: Tangible Benefits Table for CBA*

Intangible Benefits

| **Benefits** | **In-House team** | **Outsourced** |
|---|---|---|
| Cultural Alignment | Strong | Moderate |
| Turnover Risk | High | None |
| Scalability | Low | High |

| | | |
|---|---|---|
| Specialized Skills | Low | High |

*Figure 8: Intangible Benefits Table for CBA*

| PROJECTED FY-2: SENSITIVITY ANALYSIS BASED ON FY-1 | | | | |
|---|---|---|---|---|
| **SKU** | **FY 1 Baseline ED%** | **-10% Tanks (Proj. FY2 ED%)** | **+10% Tanks (Proj. FY2 ED%)** | **Notes** |
| BR-1 | 5.54% | +15.01% | –3.93% | Needs slightly more supply; higher PMD in peak months reduces shortages. Avoid lowering PMD.. |
| BR-2 | 5.41% | +14.87% | –3.96% | Similar to BR-1; a modest PMD increase smooths shortages. Large PMD reductions worsen service levels. |
| BR-3 | -9.57% | +0.69% | –19.83% | Extremely surplus-prone; reduce PMD in winter/off-peak months. Very sensitive to adjustments. |
| BR-4 | -7.86% | +1.93% | –17.65% | Volatile SKU; small PMD reductions help stabilize production. Over-increasing PMD creates deep surplus. |
| BR-5 | -2.72% | +6.45% | –11.89% | Near-balanced; slight PMD reductions help correct minor surplus. Low sensitivity. |
| BR-6 | 4.13 % | +13.72% | –5.46% | Shortage SKU; increasing PMD slightly improves alignment. Surplus risk is low |
| BR-7 | -4.59% | +13.72% | –5.46% | Mild shortage SKU; increasing PMD during peak demand helps alignment. Reducing PMD risks unmet demand. |

| | | | | |
|---|---|---|---|---|
| BR-8 | -9.57% | +4.13% | –13.31% | Tends toward surplus; reduce PMD slightly in early months. Increasing PMD pushes systems into deeper surplus. |

*Figure 9: Projected FY-2: Sensitivity analysis Based On FY-1*

| **PROJECTED FY-3: SENSITIVITY ANALYSIS BASED ON FY-2** | | | | |
|---|---|---|---|---|
| **SKU** | **FY 1 Baseline ED%** | **-10% Tanks (Proj. FY2 ED%)** | **+10% Tanks (Proj. FY2 ED%)** | **Notes** |
| BR-1 | 1.62% | +12.56% | –8.21% | BR-1 is stable; PMD changes should be minimal. Raising PMD too much causes mild surplus; lowering PMD causes shortages. |
| BR-2 | 6.78% | +16.46% | –2.15% | BR-2 consistently underproduces. Increasing PMD slightly (e.g., +1–2%) will correct shortages and stabilize service levels. |
| BR-3 | -2.08% | +7.87% | –12.29% | BR-3 is close to balanced. Lowering PMD increases shortages; raising PMD risks deeper surplus. Maintain current PMD.. |
| BR-4 | -0.49% | +9.56% | –10.53% | BR-4 is highly volatile. Small PMD shifts cause large swings; adjust monthly PMD values cautiously and in very small increments. |
| BR-5 | 4.30% | +13.87% | –2.17% | BR-5 has minor shortages. Slightly increasing PMD in high-season months helps meet demand; avoid excessive increases. |
| BR-6 | 4.21 % | +6.20% | –14.63% | BR-6 shows a persistent surplus pattern. Reduce PMD |

| | | | | |
|---|---|---|---|---|
| | | | | slightly, especially in off-peak months, to improve accuracy. |
| BR-7 | 2.56% | +6.20% | –14.63% | BR-7 is persistently surplus. Reduce PMD by 1–2% in low-season months to avoid overproduction. |
| BR-8 | -2.08% | +12.30% | –7.88% | BR-8 is slightly short. Increasing PMD marginally improves alignment; avoid large PMD increases which may create surplus. |

*Figure 10: Projected FY-2: Sensitivity analysis Based On FY-1*

*Figure 11: Methodological Approach*

FY-1 is used as the baseline because it provides the first complete and observed relationship between demand and supply for each SKU. Excess Demand (ED) is calculated to quantify how well production is aligned with actual demand. ED is defined mathematically as:

Excess Demand Formula = Demand - Supply / Demand x 100

A positive ED% indicates a shortage (demand exceeds supply), while a negative ED% indicates a surplus (supply exceeds demand). This metric helps identify which SKUs require increased or decreased production emphasis when designing the Percent Monthly Demand distribution for future fiscal years.

To evaluate how adjustments to Percent Monthly Demand influence ED, the annual demand allocated to each month is proportionally increased or decreased based on the SKU's baseline ED%. This follows standard operations logic: If ED% is positive (shortage) - the SKU requires a

higher share of monthly demand allocation in FY-2. If ED% is negative (surplus) - the SKU requires a reduced monthly demand allocation in FY-2.

Mathematically, an adjustment factor is applied to the monthly demand proportion:

$$\text{Adjusted Monthly Demand} = \text{Base Monthly Demand} \times (1+\alpha)$$

Where **α** is a small adjustment (typically ±1–5%) determined by the severity of the ED%.

**Sensitivity Analysis Methodology Applied Example:**

If BR-6 showed a +8% ED in FY-1, supply met only 92% of its demand. To correct this mismatch, the Percent Monthly Demand for BR-6 in FY-2 is increased. For instance, if March originally captured 10% of BR-6's annual demand, a 2–3% upward adjustment is applied:

$$10\% \times (1+0.02) = 10.2\%$$

This proportional adjustment across all months raises the annual demand allocated for BR-6, helping bring ED closer to the target range of –10% to +10%.

Conversely, SKUs with surplus (e.g., BR-8 at –4.59% ED) receive a downward adjustment:

$$8\% \times (1-0.02) = 7.84\%$$

These adjustments ensure that total annual demand still sums to 100%, but the distribution across SKUs shifts to better match operational realities.

Finally, FY-2 becomes the updated empirical baseline. ED is recalculated after implementing the adjusted Percent Monthly Demand. A second refinement cycle is then applied to develop FY-3

allocations. This step-by-step method follows continuous improvement principles by using each year's performance to inform more accurate planning for the next.

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